

Biophysics - Expt 1

To estimate the diffusion coefficient from the blob-diffusion experiment

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Abstract

This experiment aims to estimate the diffusion coefficient by analyzing the spread of a dye blob in a medium over time. We analyze the spread using the Image-J software. The principle of diffusion is based on the random motion of particles from regions of higher concentration to regions of lower concentration, leading to an even distribution over time. By measuring the change in the area of the dye blob at different time intervals, we can calculate the diffusion coefficient using Fick's laws of diffusion.

Aim

To estimate the diffusion coefficient from the blob-diffusion experiment.

Apparatus

- Petri dish
- Water and Glycerol
- KMnO_4 solution
- Image-J software
- Pipette

Working Principle

The diffusion of the dye in the medium is governed by Fick's laws of diffusion, which describe how the concentration of a substance changes over time and space. In this experiment, we observe the spread of the KMnO_4 dye in a water-glycerol mixture, which creates a gradient that drives the diffusion process. By capturing images at regular intervals and analyzing them with Image-J, we can quantify the diffusion coefficient. The method used to calculate the diffusion coefficient is finding the FWHM (Full Width at Half Maximum) of the dye concentration profile, which provides a measure of how far the dye has spread over time. The diffusion coefficient (D) can be estimated using the relation:

$$D = \frac{\sigma_2^2 - \sigma_1^2}{6(t_2 - t_1)}$$

where σ_1 and σ_2 are the standard deviations of the dye concentration profile at times t_1 and t_2 , respectively. The FWHM is related to the standard deviation (σ) of a Gaussian distribution by the formula:

$$FWHM = 2\sqrt{2 \ln 2} \cdot \sigma \approx 2.355 \cdot \sigma$$

Experimental Procedure

1. Measure and add 10 mL of the given solution in a 60 mm petri dish.
2. Dry the bottom and set the petri dish on a flat surface, preferably something which provides contrast, like a white paper.
3. Keep two trip-boxes to the side and place your phone or camera so that the centre of the petri dish can be imaged.
4. Take 5 μL of the KMnO_4 solution in a pipette and place a drop at the centre of the petri dish.
5. Capture a video of the KMnO_4 drop diffusing in the solution.
6. Stop the video capture when the purple colour has spread to a significant portion of the petri dish, say half of the dish.
7. Calibrate the camera by placing a ruler in the frame and measuring the distance in pixels corresponding to a known length (e.g., 1 cm).
8. Analyze the video using Image-J to extract the diffusion data.

Analysis using Image-J

1. Open the video file in Image-J.
2. Extract snapshots at relevant time intervals, noting the time stamp, precise to the second at minimum.
3. Use the red channel to analyze the images.
4. Draw an ROI across the center of the dye blob.
5. Apply the same ROI to the other time point.
6. Get the intensity profile of the dye blob at each time point using the "Plot Profile" function.
7. From the profile, get the maxima, half maxima, and the FWHM.
8. From FWHM, calculate σ using the relation mentioned above.
9. Use calibration to convert from pixels to cm.
10. Calculate Diffusion Constant D using the method mentioned above.
11. Report the findings, including any sources of error and potential improvements for future experiments.

Observations and Results

KMnO₄ in Water-Glycerol Mixture - Eklavya:

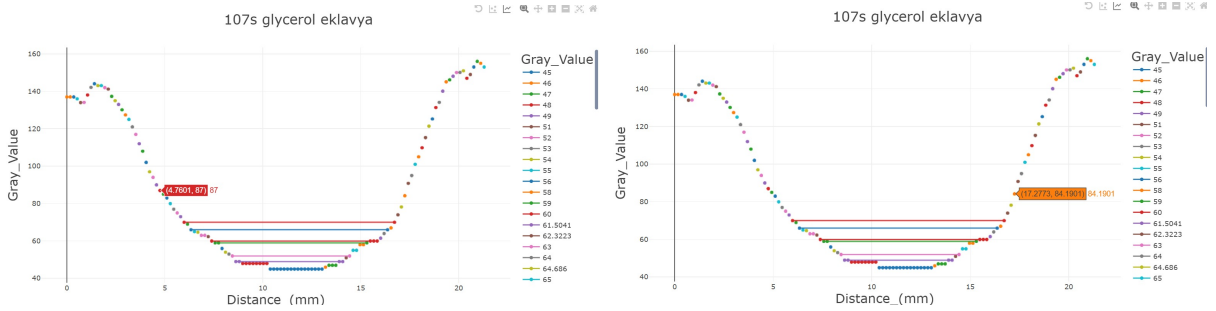


Figure 1: Graph of Intensity Profile of KMnO₄ in Water-Glycerol Mixture at 107s

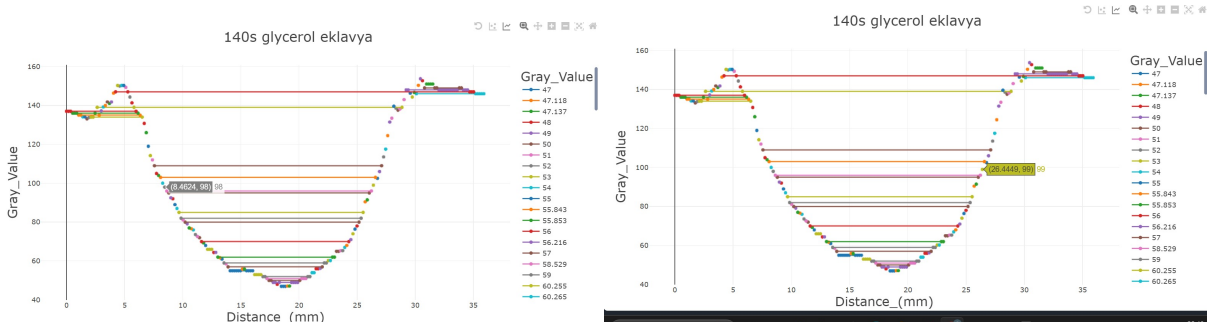


Figure 2: Graph of Intensity Profile of KMnO₄ in Water-Glycerol Mixture at 140s

We have -

$$\sigma_1 = \frac{FWHM_1}{2.355} = \frac{17.27 - 4.76}{2.355} = 5.31$$

$$\sigma_2 = \frac{FWHM_2}{2.355} = \frac{26.44 - 8.46}{2.355} = 7.63$$

$$D = \frac{\sigma_2^2 - \sigma_1^2}{6(t_2 - t_1)} = \frac{7.63^2 - 5.31^2}{6(140 - 107)} = \frac{58.22 - 28.20}{6(33)} = \frac{30.02}{198} \approx 0.1515 \text{ mm}^2/\text{s}$$

KMnO₄ in Water-Glycerol Mixture - Shrey:

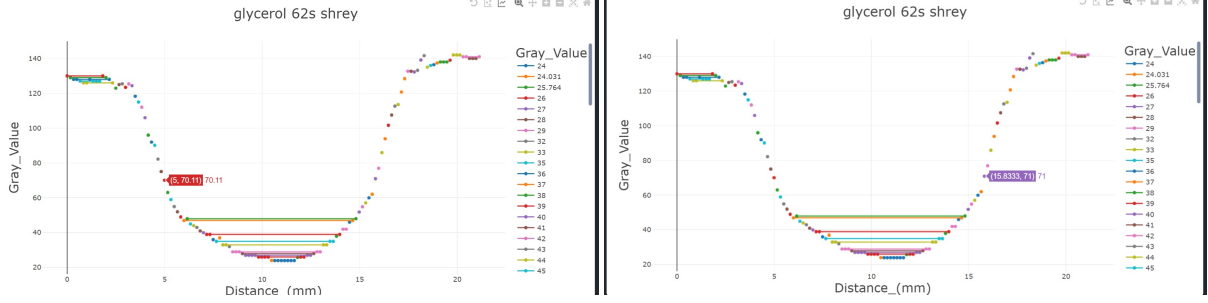


Figure 3: Graph of Intensity Profile of KMnO₄ in Water-Glycerol Mixture at 62s

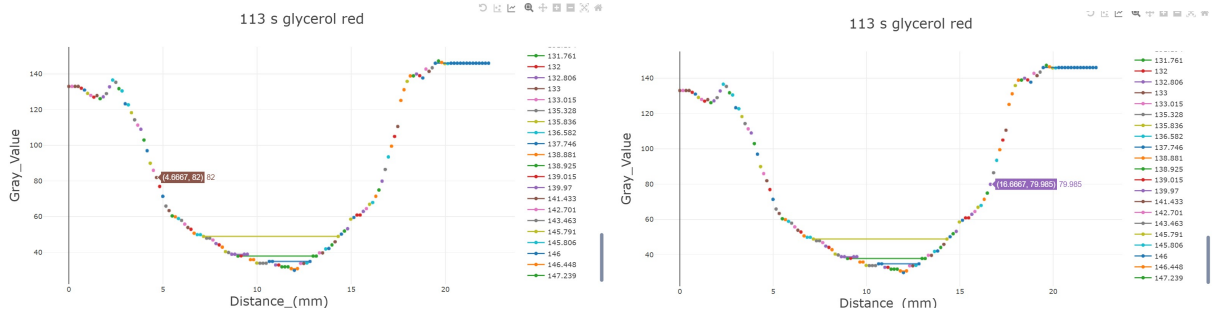


Figure 4: Graph of Intensity Profile of KMnO₄ in Water-Glycerol Mixture at 113s

We have -

$$\sigma_1 = \frac{FWHM_1}{2.355} = \frac{15.83 - 5.7}{2.355} = 4.3$$

$$\sigma_2 = \frac{FWHM_2}{2.355} = \frac{16.66 - 4.66}{2.355} = 5.1$$

$$D = \frac{\sigma_2^2 - \sigma_1^2}{6(t_2 - t_1)} = \frac{5.1^2 - 4.3^2}{6(140 - 107)} = \frac{26.01 - 18.49}{6(51)} = \frac{7.52}{306} \approx 0.02 \text{ mm}^2/\text{s}$$

KMnO₄ in Water - Eklavya:

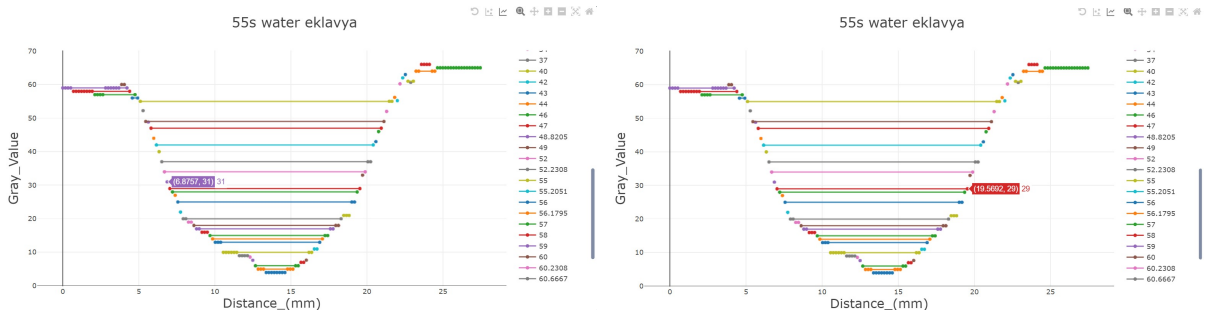


Figure 5: Graph of Intensity Profile of KMnO₄ in Water at 55s

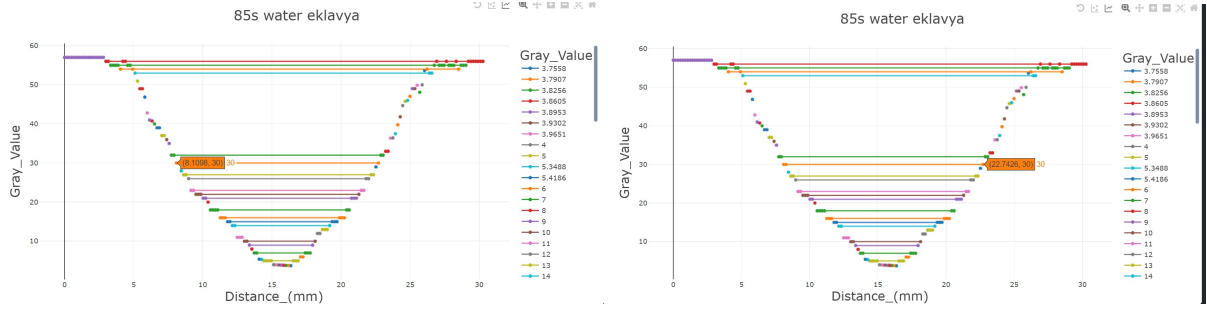


Figure 6: Graph of Intensity Profile of KMnO_4 in Water at 85s

We have -

$$\sigma_1 = \frac{FWHM_1}{2.355} = \frac{19.56 - 6.87}{2.355} = 5.38$$

$$\sigma_2 = \frac{FWHM_2}{2.355} = \frac{22.75 - 8.1}{2.355} = 6.22$$

$$D = \frac{\sigma_2^2 - \sigma_1^2}{6(t_2 - t_1)} = \frac{6.22^2 - 5.38^2}{6(85 - 55)} = \frac{38.69 - 28.94}{6(30)} = \frac{9.75}{180} \approx 0.0542 \text{ mm}^2/\text{s}$$

KMnO_4 in Water - Shrey:

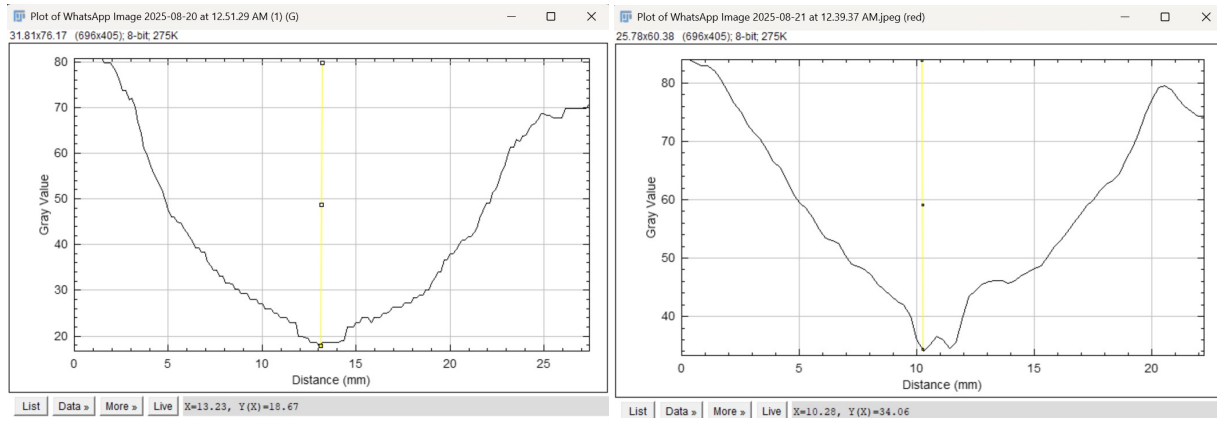


Figure 7: Graph of Intensity Profile of KMnO_4 in Water at 90s and 120s

We have -

$$\sigma_1 = \frac{FWHM_1}{2.355} = \frac{22.0 - 5.0}{2.355} = 7.21$$

$$\sigma_2 = \frac{FWHM_2}{2.355} = \frac{17 - 5}{2.355} = 5.09$$

$$D = \frac{\sigma_2^2 - \sigma_1^2}{6(t_2 - t_1)} = \frac{7.21^2 - 5.09^2}{6(120 - 90)} = \frac{51.98 - 25.90}{6(30)} = \frac{26.08}{180} \approx 0.144 \text{ mm}^2/\text{s}$$

Conclusion

The diffusion coefficients for KMnO_4 in different media were estimated using the blob-diffusion experiment and analysis with Image-J software. The calculated diffusion coefficients are as follows:

- KMnO_4 in Water-Glycerol Mixture - Eklavya: $D \approx 0.1515 \text{ mm}^2/\text{s}$
- KMnO_4 in Water-Glycerol Mixture - Shrey: $D \approx 0.02 \text{ mm}^2/\text{s}$
- KMnO_4 in Water - Eklavya: $D \approx 0.0542 \text{ mm}^2/\text{s}$
- KMnO_4 in Water - Shrey: $D \approx 0.144 \text{ mm}^2/\text{s}$